

EcoPlate AI: Project Concept Note

Deliverable 2: Concept Note (India Region & UN SDG 12 Target 12.3)

1. Executive Summary & Background

Institutional kitchens in India—such as university hostel messes (IITs, NITs) and corporate IT cafeterias in Bengaluru, Hyderabad, and Pune—prepare thousands of meals daily, yet 25-35% of prepared food ends up wasted. This inefficiency stems from an information void at disposal stations. Mess wardens log total waste weight without tracking specific dishes (Paneer Butter Masala, Basmati Rice, Dal Tadka). Consequently, kitchens over-purchase ingredients, incurring severe financial losses (INR 2 Lakh to INR 5 Lakh monthly per campus) and generating landfill methane emissions.

EcoPlate AI addresses this challenge by deploying an autonomous multi-agent AI framework powered by edge computer vision fine-tuned on regional Indian cuisine. Positioned above disposal bins, EcoPlate AI classifies and quantifies food waste in real time, translating telemetry into automated predictive supply chain adjustments.

2. System Architecture & Core AI Technologies

- Edge Computer Vision: Low-cost edge hardware (Raspberry Pi 5 / NVIDIA Jetson) running YOLOv8 fine-tuned on Indian cuisine for sub-100ms dish classification and spatial depth volume estimation.
- FastAPI Asynchronous Backend: High-performance Python backend processing vision telemetry, mapping volume to financial cost in INR, and calculating CO₂e avoided.
- React Web Dashboard: Responsive interface presenting real-time financial loss counters in INR, waste category heatmaps, and downloadable ESG compliance reports.

3. Autonomous Multi-Agent Workflow (Zero Human Intervention)

- Step 1 (Vision Agent): Scans disposal frames, tags Indian dish categories, calculates mass and monetary loss in INR.
- Step 2 (Analytics Agent): Correlates waste streams against 30-day historical trends, weather, and exam schedules.
- Step 3 (Procurement Agent): Calculates recipe batch size reductions and auto-drafts Purchase Orders (POs) in INR.

4. Measurable Impact on UN SDG 12 & FSSAI / SEBI BRSR

- Waste Volume: 40% reduction in organic waste (~140 kg saved daily per mess, 50 Metric Tons/year).
- Financial Savings: INR 1,50,000 - INR 4,50,000 saved per month in procurement spend.
- Decarbonization: ~120 Metric Tons of CO₂e avoided annually per mess.
- Compliance: 100% digital audit readiness for FSSAI Save Food Share Food & SEBI BRSR ESG disclosures.